

The Islamia University Bahawalpur

Department of Electronic Engineering

Faculty of Engineering



Complex Engineering Problem (CEP)

Artificial Intelligence (COMP-03661)

Submitted to:

Dr. Dileep Kumar

Submitted by:

Shahroz Chaudhary (F22BETEN1M01017)

Salamat Ali (F22BETEN1M01019)

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1. Title

IoT-Enabled Edge-AI Smart Lighting Control System

2. Equipment Used

ESP32 microcontroller, LDR sensor, temperature sensor, relay module, TFT LCD display, and 5V DC power adapter.

3. Introduction

This project focuses on the development of an IoT-enabled Edge-AI smart lighting control system designed to optimize energy consumption through intelligent decision-making. By performing machine learning inference directly on the ESP32, the system reduces latency and dependency on cloud resources.

4. Methodology

The proposed system acquires real-time data from light intensity and temperature sensors interfaced with the ESP32 microcontroller. The sensor readings are normalized and processed locally using a trained machine learning model deployed on the device. Based on the predicted lighting condition, the relay module is actuated to control the lighting load. System parameters and operational status are continuously uploaded to the IoT dashboard for monitoring and analysis.

5. Result

Experimental evaluation shows that the Edge-AI-based lighting control system operates reliably under varying environmental conditions. The AI-based approach demonstrates improved accuracy and energy efficiency compared to conventional threshold-based control.

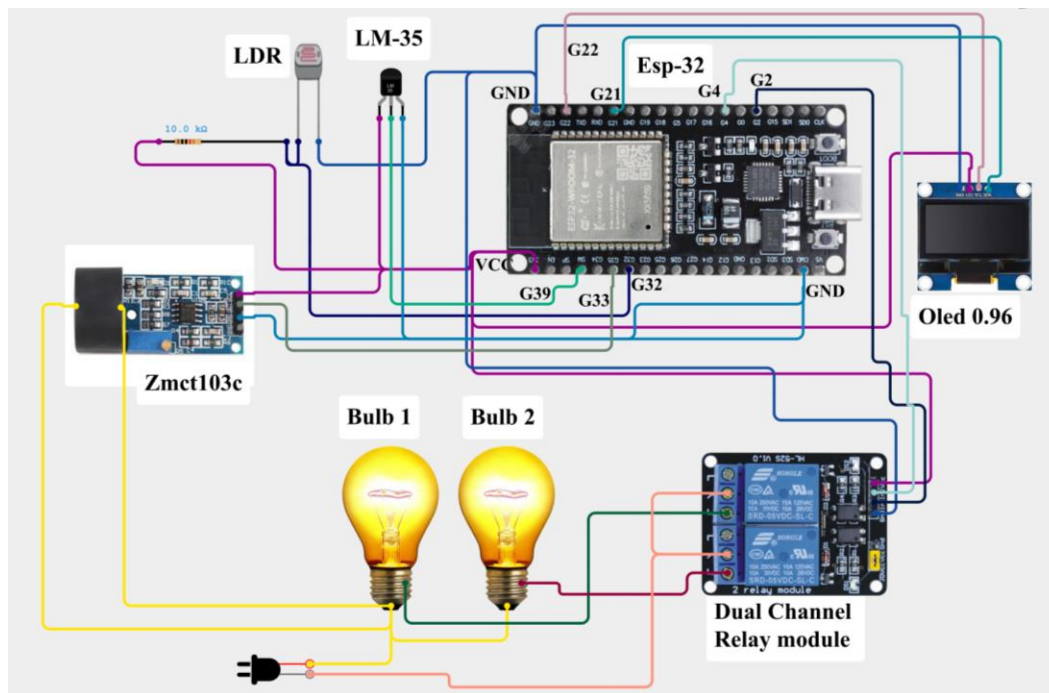
6. Conclusion

The implementation confirms that integrating edge-based machine learning with IoT systems can significantly enhance energy efficiency and system intelligence. The proposed design is suitable for real-world smart lighting applications.

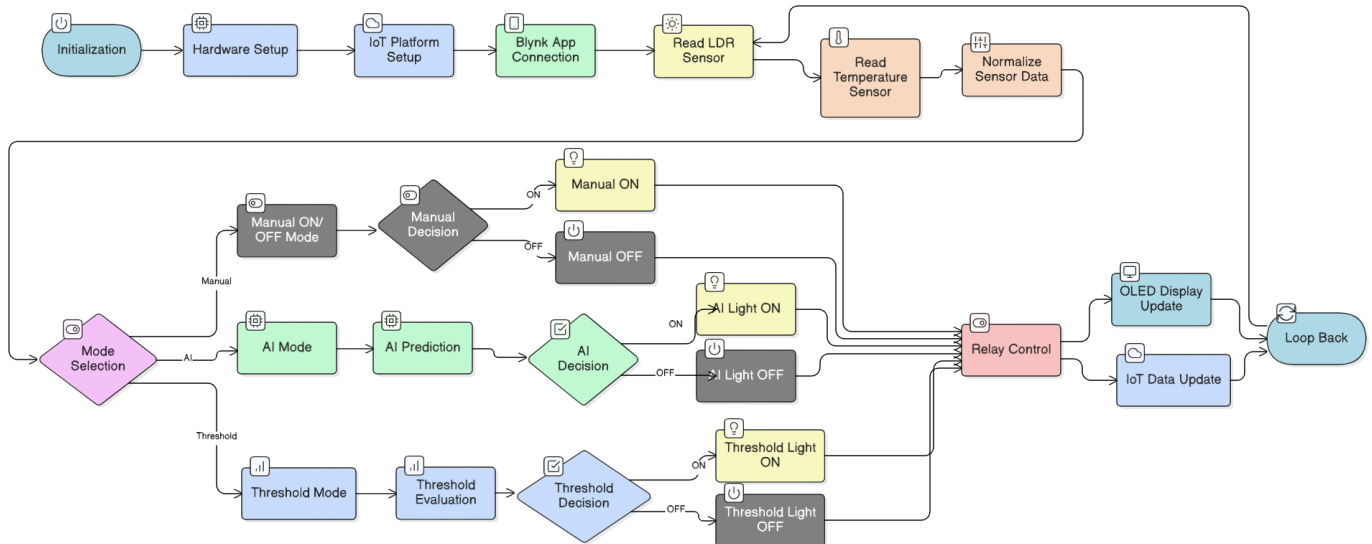
7. System Components and Hardware

Component Name	Specification	Role in System	Approx. Price
ESP32	Dual-core, Wi-Fi	Edge controller	PKR 1400
LDR Sensor	0–2000 lux	Ambient light sensing	PKR 150
LM-35	-55–150 °C	Context sensing	PKR 300
Dual Channel Relay Module	10A, 220V	Lighting control	PKR 400
Zmct103c	5A	For Power	PKR 375
Oled	0.96 inch	User feedback	PKR 490
Resistance	10kohm	Voltage divider	PKR 5
		Total	PKR 3090

8. Schematic Diagram and Pin-Level Connectivity



9. Logical Flow and Control Strategy Representation



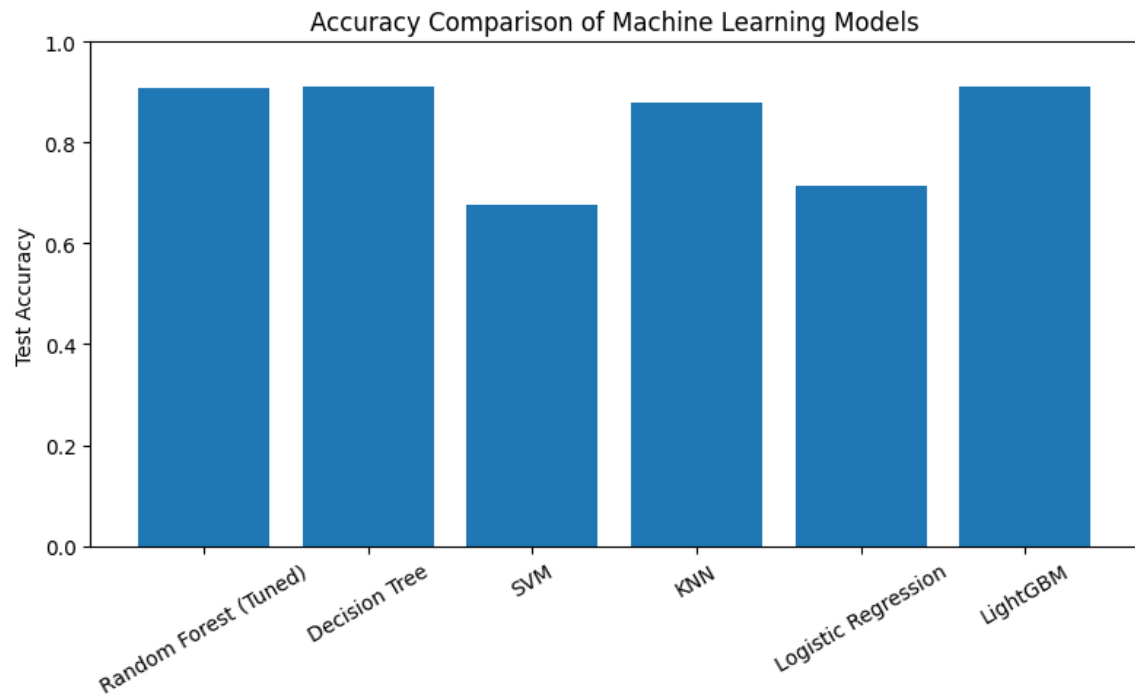
10. Data Collection and Real-World Scenario Mapping

Scenario	LDR	Temp	AI Prediction	Light Action	Folder Path
Early morning (low light)	300	15 °C	Dim (0)	☀️ Light ON	\lidar_test\Dataset\data_morning
Late morning (bright)	1200	22 °C	Bright (1)	💡 OFF	\lidar_test\Dataset\after_sunset_20mint.csv
Noon (glare sunlight)	1650	28 °C	Glare (2)	💡 OFF	\lidar_test\Dataset\data_noon
Cloudy outdoor	1100	19 °C	Bright (1)	💡 OFF	\lidar_test\Dataset\data_noon
Indoor lights off	400	17 °C	Dim (0)	☀️ ON	\lidar_test\Dataset\indoornoon
Indoor lights on	900	20 °C	Bright (1)	💡 OFF	\lidar_test\Dataset\indoornoon
Evening sunset	450	16 °C	Dim (0)	☀️ ON	\lidar_test\Dataset\noondata

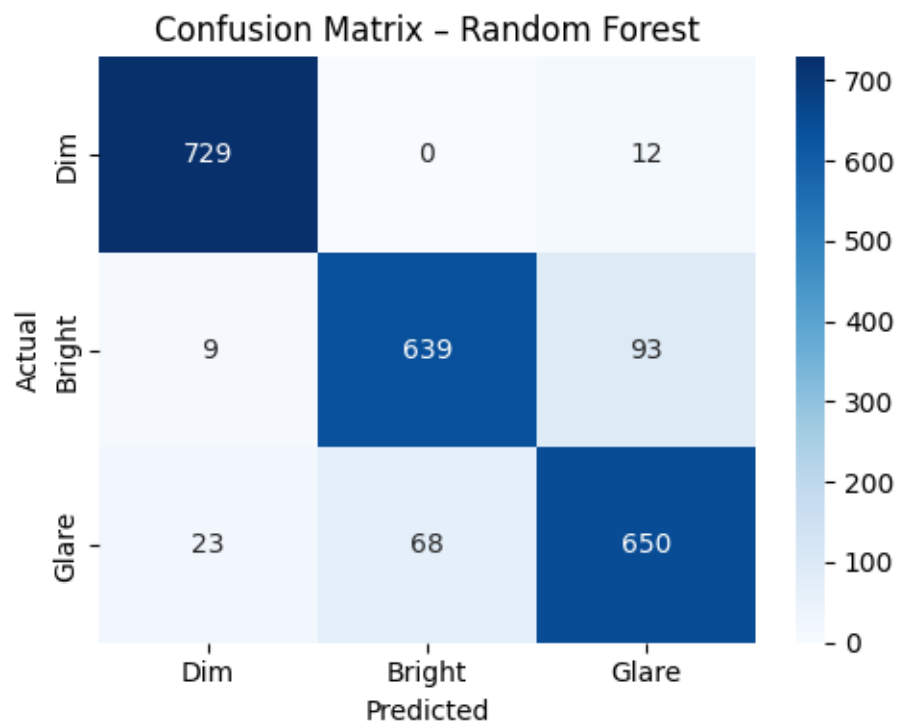
Night (streetlight)	280	14 °C	Dim (0)	☀️ ON	\lidar_test\Dataset\night_ambient
High-temperature test	1100	30 °C	Glare (2)	💡 OFF	\lidar_test\Dataset\sunset_time
Fan/cool air test	950	15 °C	Bright (1)	💡 OFF	\lidar_test\Dataset\sunset_time

11. Machine Learning Model Selection and Tuning

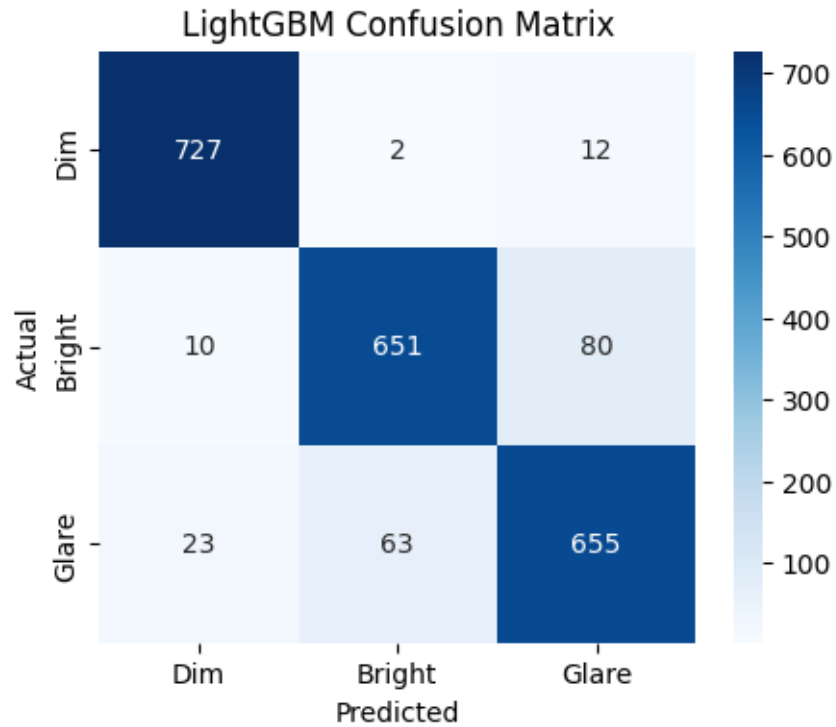
MODEL	ACCURACY	PRECISION	RECALL	F1-SCORE	MCC
RANDOM FOREST	0.91318	0.912595	0.91318	0.912809	0.869844
DECISION TREE	0.910031	0.909365	0.910031	0.90953	0.865206
GRADIENT BOOSTING	0.907332	0.907758	0.907332	0.907055	0.86146
KNN	0.878093	0.877397	0.878093	0.87771	0.817171
ADABOOST	0.764283	0.781364	0.764283	0.757254	0.661563
QDA	0.724696	0.721882	0.724696	0.713133	0.595495
LOGISTIC REGRESSION	0.7148	0.713953	0.7148	0.713848	0.572688
NAIVE BAYES	0.71345	0.709095	0.71345	0.702607	0.577369
SVM	0.677463	0.75171	0.677463	0.685494	0.54088



Confusion Matrix of Random Forest



Confusion Matrix of Light GBM

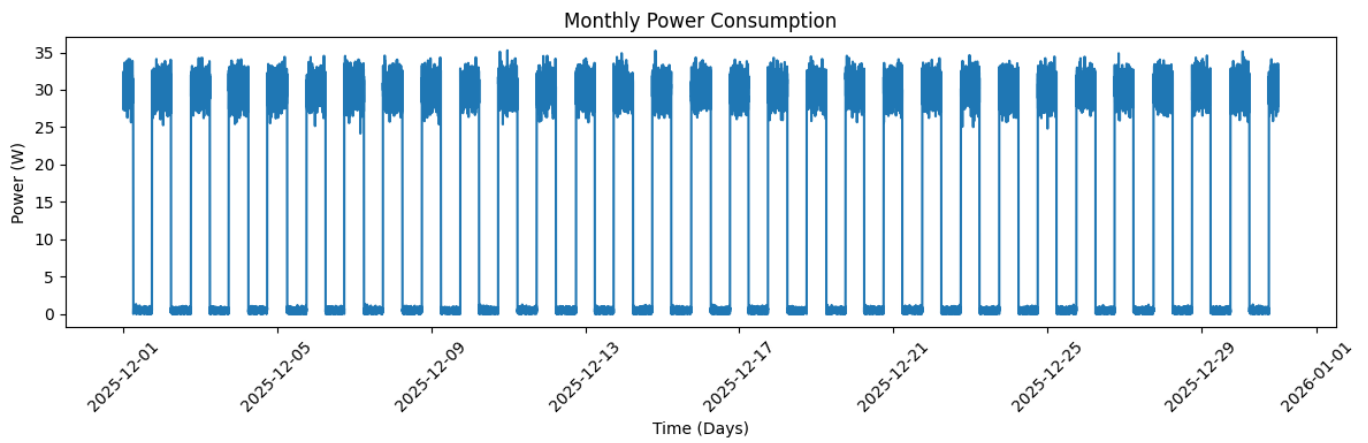
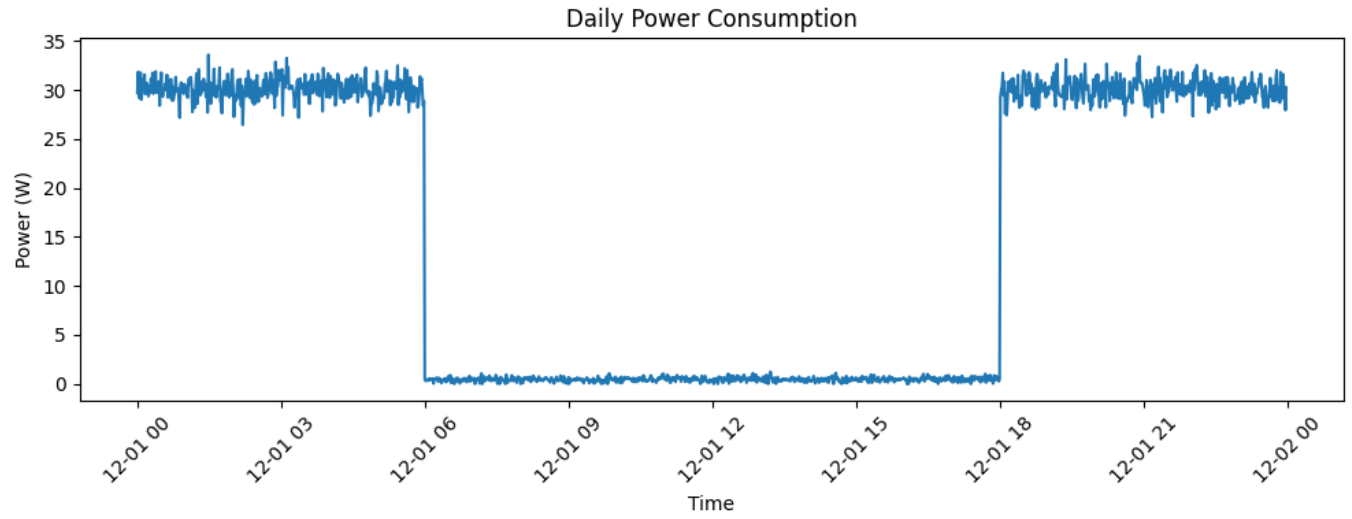


12. Energy Consumption Comparison

Mode	Per hour power Consumption	Per Day Power Consumption	Unit consumes in Month
Threshold	29.04 Wh	$29.04 \times 11 = 319.44$	9.583
Ai	29.04 Wh	$29.04 \times 10 = 290.4$	8.712

Realtime Test:

Dim light On	In Shadow	05:37:25 PM
Fully Light On	In Shadow	05:42:36 PM
Dim Light On	In open area	05:45:02 PM
Fully Light On	In open area	05:49:45 PM
From Fully to Dim	In Shadow	6:47:25 AM
From Dim to Off	In Shadow	6:57:33 AM
From Fully to Dim	In open area	6:27:33 AM
From Dim to Off	In open area	6:32:33 AM



13. Model Size and Edge Optimization Analysis

n_estimators	max_depth	min_samples_split	min_samples_leaf	Train Accuracy	Test Accuracy	Model Size (KB)
50	Nan	10	2	0.923838	0.916329	576.5869
50	20	10	2	0.923838	0.916329	576.1943
100	Nan	10	2	0.924063	0.916329	1152.382
75	Nan	10	2	0.923726	0.915879	864.5781
50	15	10	4	0.922713	0.915879	513.3213
111	20	10	2	0.923951	0.915879	1276.437
100	20	10	2	0.923951	0.915879	1151.958
111	Nan	10	2	0.923951	0.915879	1277.691
50	20	10	1	0.925188	0.91543	591.1699

14. IoT Dashboard Data Representation

